ABHIJEET GUPTA

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ML / AI Engineer | CS PhD Candidate | PyTorch | LLM Evaluation | RL | Computer Vision

SUMMARY

CS PhD candidate graduating May 2027 with research and industry experience in controlled model evaluation, reinforcement learning, computer vision, production data systems, and multi-GPU inference. Public repositories distinguish published claims, reconstructed implementations, and newly reproduced results.

TECHNICAL SKILLS

ML: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Gymnasium, OpenCV, MediaPipe, CNNs, LSTMs, reinforcement learning, multilabel learning

Research engineering: Python, NumPy, pandas, SciPy, statsmodels, Slurm/HPC, multi-GPU inference, Docker, Kubernetes, GitHub Actions, deterministic experiments

EXPERIENCE

Graduate Research Assistant | University of Dayton **| Aug 2023 - Present**

- Builds controlled ML evaluation pipelines spanning adversarial reinforcement learning, computer vision, and reasoning-model analysis; coordinates reproducible Slurm and multi-GPU experiments.
- Co-authored peer-reviewed ML/CV research and mentors junior researchers on experiment design, PyTorch implementation, statistical validation, and publication artifacts.

Data & Systems Engineering Intern | Dayton Phoenix Group **| Jan 2025 - Present**

- Built Python data pipelines and KPI reporting for manufacturing systems; applied statistical process control and anomaly analysis to production metrics.

Software Engineer | Walmart Global Tech via Kitestring **| Apr 2023 - Aug 2023**

- Built Docker/Kubernetes CI/CD workflows for a Node.js point-of-sale application and Python validation tooling for migration of more than one million records to Azure Cosmos DB.

ML / Product Intern | Walmart **| Jun 2022 - Aug 2022**

- Developed a Random Forest supply-chain classifier and integrated heterogeneous operational data sources with leakage-aware validation.

SELECTED RESEARCH PROJECTS

Reasoning Model Failure Analysis | Private research repository

- Designed controlled evaluations across six reasoning models to separate reasoning-length effects from forced re-entry interventions; original artifacts remain private.
- Implemented multi-GPU inference and structured behavioral/mechanistic result capture. No private dataset or participant material is published.

Adversarial RL Feature Selection | Public reference reconstruction

- Implemented the paper's Lunar Lander/Bipedal Walker dimensions, Gaussian/uniform imposter injection, entropy/joint-entropy/KL descriptors, and Naive/RF/KNN/SVM detectors.
- Paper-reported best accuracies are 97.05% (Lunar Lander, RF/entropy) and 97.77% (Bipedal Walker, SVM/entropy); local fixture results are explicitly separate.

Mouse Brain Cell Segmentation | Public reference reconstruction

- Implemented a six-class COCO instance-segmentation contract and exact configurations for Mask R-CNN, CenterMask2, YOLACT++, Mask2Former, MaskDINO, and FastInst.
- Paper-reported Mask2Former AP is 30.2 and AR10 is 34.1; private-dataset training remains NOT_RUN and synthetic evaluator metrics are not presented as biological results.

Virtual Yoga Instructor | Public reference reconstruction

- Implemented MediaPipe's 33-landmark schema, eight bilateral angles, tolerance scoring, 16 directional prompts, marker coordinates, and acknowledged TCP/JSON messages.
- Measured 0.019 ms geometry-only scoring latency; camera, MediaPipe, network, and Unity latency remain NOT_RUN and are not implied by this result.

Multi-Output Career Prediction | Public reference reconstruction

- Implemented the paper's 26-to-11 entropy feature selection, 6-class domain and 8-class position outputs, exact 300/120 split, and all six multiclass-multioutput classifiers.
- Paper-reported Random Forest accuracy is 91.21% for domain and 95.97% for position; personal LinkedIn-derived records are not distributed.

GROVE Construction Safety Evaluation | Public research companion

- Packaged permitted GROVE evidence with component ablations, paired statistical tests, threshold sensitivity, failure attribution, and machine-readable claim traceability.

EDUCATION

PhD, Computer Science | University of Dayton | Expected May 2027

Research: LLM evaluation and interpretability, reinforcement learning, computer vision, and reproducible machine learning.

MS, Computer Science | University of Texas at Arlington | 2022

BE, Computer Science and Engineering | RTM Nagpur University | 2019